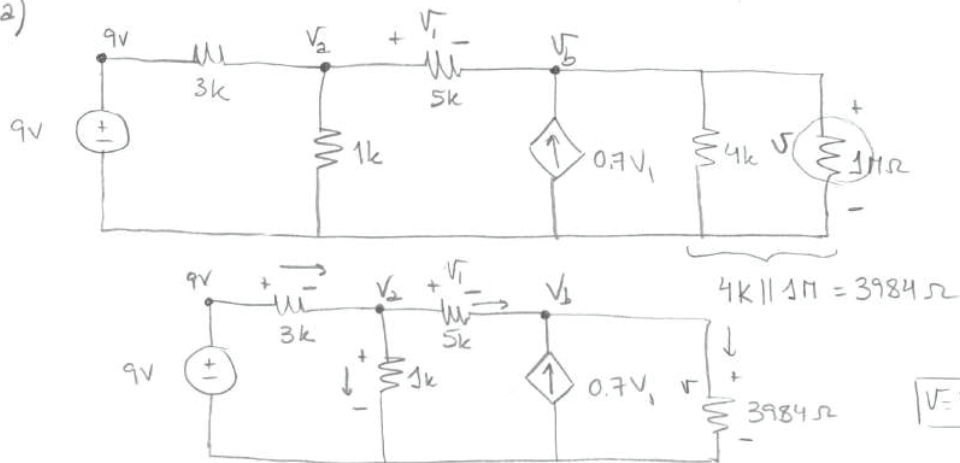


# QUESTION 1

2)



\* KCL @ node 2:

$$\frac{9 - V_2}{3k} - \frac{V_2}{1k} - \frac{V_2 - V_b}{5k} = 0$$

↓ (x15)

$$45 - 5V_2 - 15V_2 - 3V_2 + 3V_b = 0 \Rightarrow -23V_2 + 3V_b = -45$$

\* KCL @ node b:

$$\frac{V_2 - V_b}{5000} + 0.7V_1 - \frac{V_b}{3984} = 0, \text{ where } V_1 = V_2 - V_b$$

↓ (x5000)

$$V_2 - V_b + 3500(V_2 - V_b) - 1.255V_b = 0$$

$$V_2 - V_b + 3500V_2 - 3500V_b - 1.255V_b = 0$$

$$3501V_2 - 3502.255V_b = 0$$

Solve for  $V_b$ :

$$-23V_2 + 3V_b = -45 \quad (\times 3501)$$

$$3501V_2 - 3502.255V_b = 0 \quad (\times 23)$$

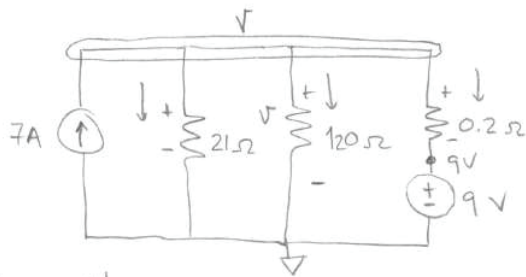
$$\begin{cases} -80523V_2 + 10503V_b = -157545 \\ 80523V_2 - 80551.87V_b = 0 \end{cases}$$

$$-70048.87V_b = -157545$$

$$V_b = V = 2.25V$$

# QUESTION 1

b)



Switch in position 2

i) Using nodal:

$$7 - \frac{V}{21} - \frac{V}{120} - \frac{V-9}{0.2} = 0$$

$$17640 - 120V - 21V - 12600(V-9) = 0$$

$$17640 + 113400 = (120 + 21 + 12600)V$$

$$V = \frac{131040}{12741} = 10.28V$$

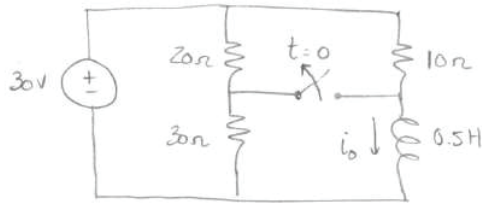
(ii)

$$V_{0.2\Omega} = V - 9 = 10.28 - 9 = 1.28V$$

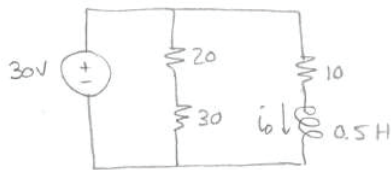
$$P_{0.2\Omega} = \frac{V^2}{R} = \frac{1.28^2}{0.2} = 8.192W$$

## QUESTION 2

2)



i) For  $t \geq 0 \Rightarrow$  Switch is open

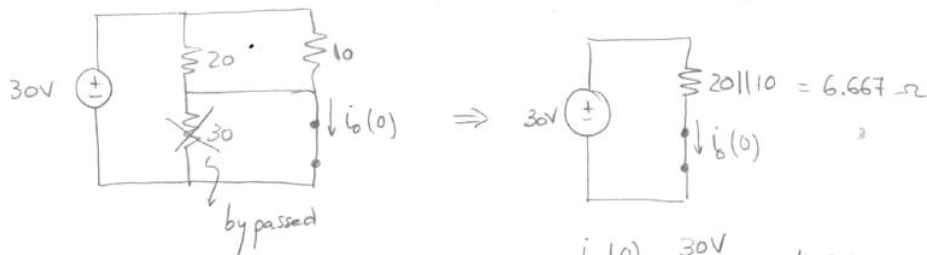


Step response

$$i_o(t) = i_o(\infty) + [i_o(0) - i_o(\infty)] e^{-t/\tau}$$

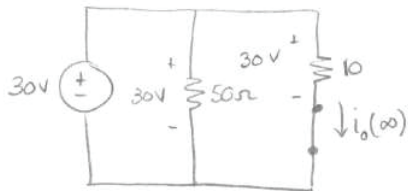
where  $\tau = L/R$

\* Find  $i_o(0)$  for  $t < 0$  (inductor is short circuit in steady state)



$$i_o(0) = \frac{30V}{6.667} = 4.5A$$

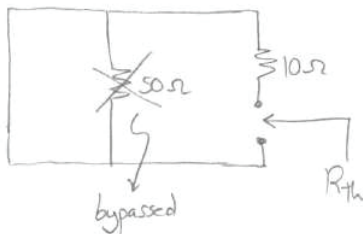
\* Find  $i_o(\infty)$  for  $t > 0$  (inductor is short circuit in steady state)



$$i_o(\infty) = \frac{30V}{10\Omega} = 3A$$

due to parallel

\* Find  $\tau$  for  $t > 0$



$$R_{th} = 10\Omega \Rightarrow \tau = \frac{L}{R_{th}} = \frac{0.5}{10} = 0.05s$$

\* Substitute:

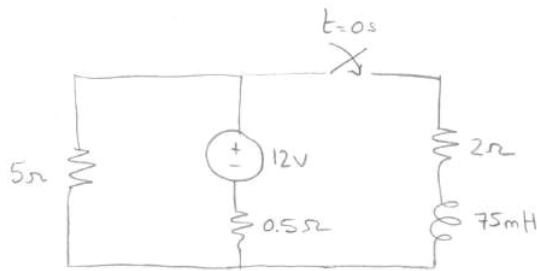
$$i_o(t) = 3 + [4.5 - 3] e^{-t/0.05} = 3 + 1.5 e^{-20t} A$$

ii)

$$i_o(0.04s) = 3 + 1.5 e^{-20 \times 0.04} = 3.674 A$$

## QUESTION 2

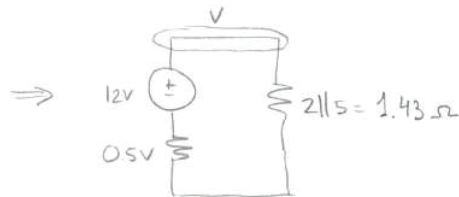
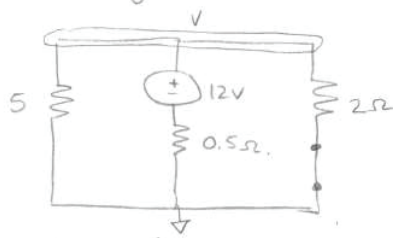
b)



i)

$$P_{\text{headlights}} = P_{\text{resistor}} + P_{\text{inductor}}$$

In DC steady state, inductor behaves as short circuit and  $P_{\text{inductor}} = 0W$   
(since  $V_{\text{inductor}} = 0V$ )

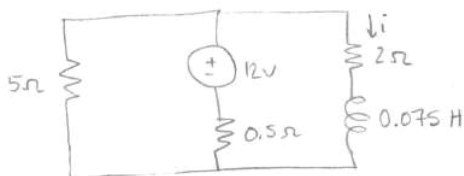


$$V = 12 \frac{1.43}{1.43 + 0.5} = 8.89V$$

↑  
voltage divider

$$P_{\text{headlight}} = P_{\text{resistor}} = \frac{V^2}{2} = \frac{8.89^2}{2} = 39.52W$$

ii)



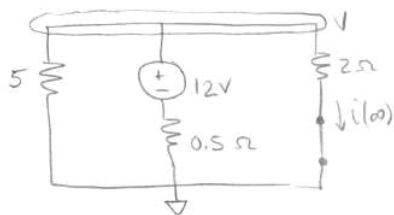
Step response:

$$i(t) = i(\infty) + [i(0) - i(\infty)] e^{-t/\tau}$$

where  $\tau = L/R$

\*  $i(0) = 0A$

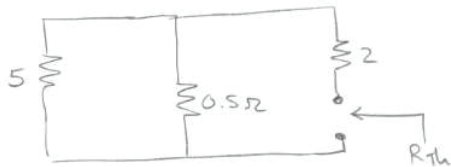
\* Calculate  $i(\infty)$  for  $t > 0$  (inductor is short circuit in steady state)



From i) we know that  $V = 8.89V$

$$\text{so } i(\infty) = \frac{V}{2} = \frac{8.89}{2} = 4.445A$$

\* Calculate  $\tau$  for  $t > 0$



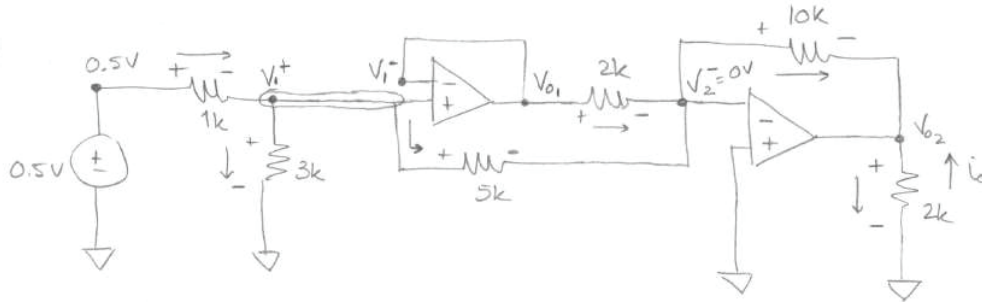
$$R_{th} = (5 \parallel 0.5) + 2 = 2.45\Omega$$

$$\tau = \frac{L}{R_{th}} = \frac{0.075}{2.45} = 0.031s$$

\* Substitute:  $i(t) = 4.445 + [0 - 4.445] e^{-t/0.031} = 4.445 (1 - e^{-32.73t}) A$

### QUESTION 3

2)



$$\text{* KCL @ } V_1^+ : \frac{0.5 - V_1^+}{1k} - \frac{V_1^+}{3k} - \frac{V_1^+ - 0}{5k} = 0$$

$$7.5 - 15V_1^+ - 5V_1^+ - 3V_1^+ = 0 \Rightarrow 23V_1^+ = 7.5 \Rightarrow V_1^+ = \frac{7.5}{23} = 0.326V$$

Op Amp 1 is a voltage follower, so  $V_{o1} = V_1^- = V_1^+ = 0.326V$

$$\text{* KCL @ } V_2 : \frac{V_1^+ - 0}{5k} + \frac{V_{o1} - 0}{2k} - \frac{0 - V_{o2}}{10k} = 0, \text{ where } V_1^+ = V_{o1} = 0.326V$$

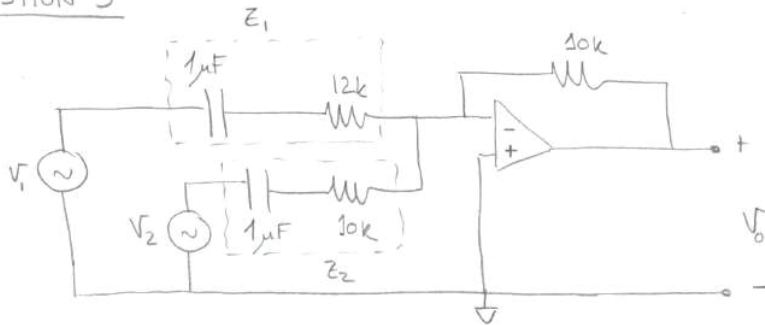
$$\frac{0.326}{5} + \frac{0.326}{2} + \frac{V_{o2}}{10} = 0 \quad \times 10$$

$$0.652 + 1.63 + V_{o2} = 0 \Rightarrow V_{o2} = -1.63 - 0.652 = -2.282V$$

$$\boxed{i_0 = -\frac{V_{o2}}{2k} = \frac{2.282}{2k} = 1.141 \text{ mA}}$$

### QUESTION 3

b)



$$|V_1| = 240 \text{ mV}$$

$$|V_2| = 210 \text{ mV}$$

$$f = 60 \text{ Hz}$$

i) First, convert to phasor domain:

$$V_1 = 0.24 \cos(120\pi t) \Rightarrow V_1 = 0.24 \angle 0^\circ \text{ V}$$

$$V_2 = 0.21 \cos(120\pi t) \Rightarrow V_2 = 0.21 \angle 0^\circ \text{ V}$$

$$Z_1 = Z_2 = -j \frac{1}{\omega C} = -j \frac{1}{120\pi \times 10^{-6}} = -j 2652.58 \Omega$$

Then, calculate  $V_0$ :

$$\text{Summing amplifier: } V_0 = - \left( \frac{10k}{Z_1} V_1 + \frac{10k}{Z_2} V_2 \right)$$

$$Z_1 = 12000 - j 2652.58$$

$$Z_2 = 10000 - j 2652.58$$

$$\begin{aligned} \boxed{V_0} &= - \left( \frac{10000}{12000 - j 2652.58} V_1 + \frac{10000}{10000 - j 2652.58} V_2 \right) = \\ &= - \left( \frac{10000}{12289.68 \angle -12.465^\circ} V_1 + \frac{10000}{10345.32 \angle -14.86^\circ} V_2 \right) = \\ &= - \left( 0.814 \angle 12.465^\circ V_1 + 0.967 \angle 14.86^\circ V_2 \right) \text{ V} \end{aligned}$$

$$\begin{aligned} \text{ii) } V_0 &= - \left( 0.814 \angle 12.465^\circ \times 0.24 \angle 0^\circ + 0.967 \angle 14.86^\circ \times 0.21 \angle 0^\circ \right) = \\ &= - \left( 0.195 \angle 12.465^\circ + 0.205 \angle 14.86^\circ \right) = - \left( 0.19 + j 0.042 + 0.198 + j 0.0526 \right) = \\ &= -0.388 - j 0.0946 = 0.4 \angle -166.3^\circ \text{ V} \end{aligned}$$

$$\boxed{V_0(t) = 0.4 \cos(120\pi t - 166.3^\circ) \text{ V}}$$

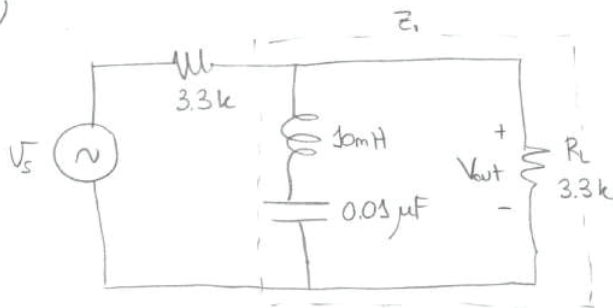
iii) Calculate voltage across  $R_1 = 12k\Omega$

$$V_{Z_1} = V_1 - 0 = V_1 \Rightarrow V_{R_1} = V_1 \frac{12k}{Z_1} = 0.24 \frac{12000}{12000 - j 2652.58} = \frac{2880}{12289.68 \angle -12.465^\circ} = 0.234 \angle 12.465^\circ \text{ V}$$

$$\boxed{P_{R_1} = \frac{1}{2} \frac{|V_{R_1}|^2}{R_1} = \frac{1}{2} \frac{0.234^2}{12000} = 2.28 \mu\text{W}}$$

# QUESTION 4

2)



i) First, convert to phasor domain

$$V_s = 1 \cos(\omega t) \Rightarrow V_s = 1 \angle 0^\circ \text{ V}$$

$$Z_L = j\omega L = j0.01\omega \text{ } \Omega$$

$$Z_C = -j \frac{1}{\omega C} = -j \frac{10^8}{\omega} \text{ } \Omega$$

Then, calculate  $V_{out}$  as a function of  $\omega$ :

$$V_{out} = V_s \frac{Z_1}{3300 + Z_1} \quad \text{where } Z_1 = 3300 \parallel \left( j0.01\omega - j \frac{10^8}{\omega} \right)$$

$$\text{voltage divider}$$

$$= \frac{3300 \times j \left( 0.01\omega - \frac{10^8}{\omega} \right)}{3300 + j \left( 0.01\omega - \frac{10^8}{\omega} \right)}$$

$$V_{out} = \frac{j \left( 33\omega - \frac{33 \times 10^{10}}{\omega} \right)}{3300 + j \left( 0.01\omega - \frac{10^8}{\omega} \right)} = \frac{j \left( 33\omega - \frac{33 \times 10^{10}}{\omega} \right)}{3300^2 + j^2 \left( 33\omega - \frac{33 \times 10^{10}}{\omega} \right)^2}$$

$$= \frac{j \left( 33\omega - \frac{33 \times 10^{10}}{\omega} \right)}{3300^2 - \left( 33\omega - \frac{33 \times 10^{10}}{\omega} \right)^2}$$

ii)  $Z_{eq} = 3300 + \left[ \left( j0.01\omega - j \frac{10^8}{\omega} \right) \parallel 3300 \right]$

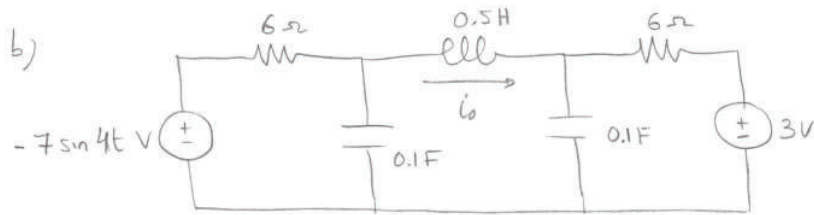
for  $Z_{eq}$  to be purely resistive  $Z_L$  and  $Z_C$  need to cancel out

$$0.01\omega = \frac{10^8}{\omega} \Rightarrow \omega^2 = \frac{10^8}{0.01} = 10^{10} \Rightarrow \omega = \sqrt{10^{10}} = 10^5 \text{ rad/s}$$

iii)  $V_{out} @ 10^5 \text{ rad/s} \Rightarrow V_{out} = \frac{j \left( 33 \times 10^5 - \frac{33 \times 10^{10}}{10^5} \right)}{3300^2 + j^2 \left( 66 \times 10^5 - \frac{66 \times 10^{10}}{10^5} \right)^2} = \frac{0}{3300^2} = 0 \text{ V}$

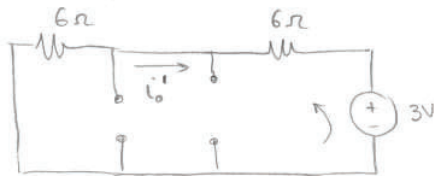
If the frequency of the interfering signal is equal (or close) to the one calculated in (ii), the amplitude of the output becomes 0 (or close), which means that the interference is filtered/removed (or attenuated).

# QUESTION 4



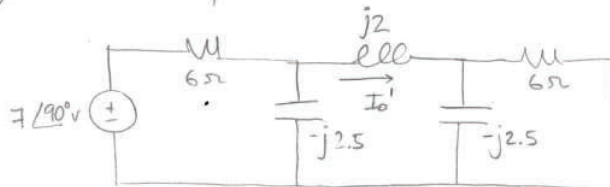
Since the 2 sources have different frequency, we need to use superposition

1) Contribution of DC source:



$$i_o' = -\frac{3V}{12\Omega} = -0.25A$$

2) Contribution of AC source:



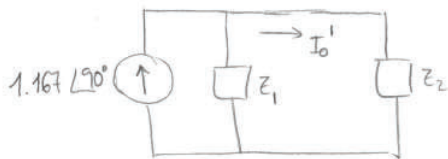
First, convert to phasor domain

$$v = -7\sin(4t) = 7\cos(4t + 90^\circ) \Rightarrow V = 7 \angle 90^\circ V$$

$$Z_C = -j \frac{1}{\omega C} = -j \frac{1}{4 \times 0.1} = -j2.5 \Omega$$

$$Z_L = j\omega L = j4 \times 0.5 = j2 \Omega$$

Then, find  $I_o'$



$$Z_1 = 6 \parallel (-j2.5)$$

$$Z_2 = j2 + [6 \parallel (-j2.5)] = j2 + Z_1$$

$$Z_1 = 6 \parallel (-j2.5) = \frac{6 \times (-j2.5)}{6 - j2.5} = \frac{15 \angle -90^\circ}{6.5 \angle -22.62^\circ} = 2.31 \angle -67.38^\circ = 0.89 - j2.13 \Omega$$

$$Z_2 = j2 + Z_1 = 0.89 - j0.13 \Omega$$

$$I_o' = \frac{1.167 \angle 90^\circ \times Z_1}{Z_1 + Z_2} = \frac{1.167 \angle 90^\circ \times 2.31 \angle -67.38^\circ}{0.89 - j2.13 + 0.89 - j0.13} = \frac{2.696 \angle 22.62^\circ}{1.78 - j2.26} = \frac{2.696 \angle 22.62^\circ}{2.877 \angle -51.77^\circ} = 0.94 \angle 74.39^\circ A$$

3) Add contributions in the time domain

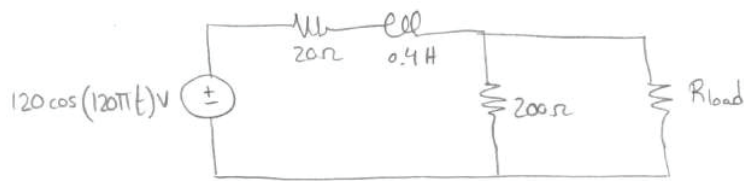
$$i_o' = -0.25A$$

$$I_o'' = 0.94 \angle 74.39^\circ \Rightarrow i_o'' = 0.94 \cos(4t + 74.39^\circ) A$$

$$i_o = i_o' + i_o'' = -0.25 + 0.94 \cos(4t + 74.39^\circ) A$$



# QUESTION 5

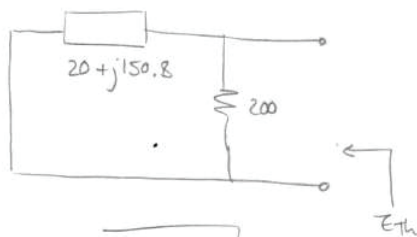


First, convert to phasor domain

$$V_{in} = 120 \cos(120\pi t) \Rightarrow V_{in} = 120 \angle 0^\circ \text{ V}$$

$$Z_L = j\omega L = j120\pi \times 0.4 = j150.8 \Omega$$

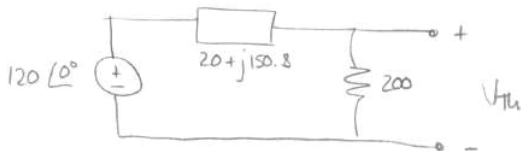
i) Calculate  $R_{load} = |Z_{Th}|$  (method 1)



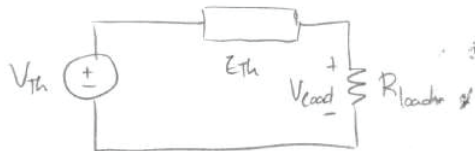
$$R_{load} = |Z_{Th}| = 114.07 \Omega$$

$$\begin{aligned} Z_{Th} &= 200 \parallel (20 + j150.8) = \frac{200(20 + j150.8)}{200 + 20 + j150.8} = \\ &= \frac{4000 + j30160}{220 + j150.8} = \frac{30424.1 \angle 82.45^\circ}{266.72 \angle 34.43^\circ} = \\ &= 114.07 \angle 48.02^\circ \Omega = 76.3 + j84.8 \Omega \end{aligned}$$

ii) To calculate  $P_{max}$ , calculate Thevenin equivalent first:



$$V_{Th} = \underset{\substack{\uparrow \\ \text{voltage} \\ \text{divider}}}{\frac{120 \times 200}{200 + 20 + j150.8}} = \frac{24000}{266.72 \angle 34.43^\circ} = 89.98 \angle -34.43^\circ$$



$$V_{load} = \underset{\substack{\uparrow \\ \text{voltage} \\ \text{divider}}}{89.98 \angle -34.43^\circ \frac{114.07}{114.07 + 76.3 + j84.8}} =$$

$$= \frac{10264.25 \angle -34.43^\circ}{190.37 + j84.8} = \frac{10264.25 \angle -34.43^\circ}{208.4 \angle 24.01^\circ}$$

$$= 49.25 \angle -58.44^\circ \text{ V}$$

$$P_{max} = \frac{1}{2} \frac{|V_{load}|^2}{R_{load}} = \frac{1}{2} \frac{49.25^2}{114.07} = 10.63 \text{ W}$$